# Topology Assignment Solutions

Topology: MAT307/M405/MAT803

Problem Sets 1–8

Submitted by **B23**

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## Contributors

| Student | Questions solved |
|---|---|
| Aaditya | PS4: Q5–8 |
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## Coverage

Solved problems are shown in **green**; pending ones in grey.

| Set | Problems | Solved |
|---|---|---|
| PS1 | 1 2 3 4 5 **6 7 8 9 10** 11 12 13 14 15 16 17 | 5/17 |
| PS2 | 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 | 0/15 |
| PS3 | **1 2 3 4 5 6 7 8 9 10** | 10/10 |
| PS4 | 1 2 3 4 **5 6 7 8** 9 10 11 | 4/11 |
| PS5 | 1 2 3 4 **5 6** 7 8 9 10 | 2/10 |
| PS6 | 1 2 3 4 5 6 7 8 9 | 0/9 |
| PS7 | 1 2 3 4 5 6 | 0/6 |
| PS8 | 1 2 3 4 5 6 7 8 9 10 | 0/10 |
| | **Total solved** | **21/88** |

# Contents

**Contributors** **1**

**Coverage** **1**

**Problem Set 1** **5**

Problem 1.1 . . . 5
Problem 1.2 . . . 5
Problem 1.3 . . . 5
Problem 1.4 . . . 5
Problem 1.5 . . . 6
Problem 1.6 – *Sahil* . . . 6
Problem 1.7 – *Sahil* . . . 7
Problem 1.8 – *Sahil* . . . 7
Problem 1.9 – *Sahil* . . . 8
Problem 1.10 – *Sahil* . . . 8
Problem 1.11 . . . 9
Problem 1.12 . . . 9
Problem 1.13 . . . 9
Problem 1.14 . . . 10
Problem 1.15 . . . 10
Problem 1.16 . . . 11
Problem 1.17 . . . 11

**Problem Set 2** **12**

Problem 2.1 . . . 12
Problem 2.2 . . . 12
Problem 2.3 . . . 12
Problem 2.4 . . . 13
Problem 2.5 . . . 13
Problem 2.6 . . . 13
Problem 2.7 . . . 13
Problem 2.8 . . . 14
Problem 2.9 . . . 14
Problem 2.10 . . . 14
Problem 2.11 . . . 14
Problem 2.12 . . . 15
Problem 2.13 . . . 15

Problem 2.14 . . . 15
Problem 2.15 . . . 15

**Problem Set 3** **16**
Problem 3.1 – *Arunodaya* . . . 16
Problem 3.2 – *Arunodaya* . . . 16
Problem 3.3 – *Arunodaya* . . . 17
Problem 3.4 – *Arunodaya* . . . 18
Problem 3.5 – *Arunodaya* . . . 19
Problem 3.6 – *Pula Karan* . . . 20
Problem 3.7 – *Pula Karan* . . . 21
Problem 3.8 – *Pula Karan* . . . 22
Problem 3.9 – *Pula Karan* . . . 23
Problem 3.10 – *Pula Karan* . . . 24

**Problem Set 4** **25**
Problem 4.1 . . . 25
Problem 4.2 . . . 25
Problem 4.3 . . . 25
Problem 4.4 . . . 25
Problem 4.5 – *Aaditya* . . . 26
Problem 4.6 – *Aaditya* . . . 26
Problem 4.7 – *Aaditya* . . . 27
Problem 4.8 – *Aaditya* . . . 28
Problem 4.9 . . . 29
Problem 4.10 . . . 29
Problem 4.11 . . . 29

**Problem Set 5** **30**
Problem 5.1 . . . 30
Problem 5.2 . . . 30
Problem 5.3 . . . 30
Problem 5.4 . . . 30
Problem 5.5 – *Prabhu* . . . 30
Problem 5.6 – *Prabhu* . . . 31
Problem 5.7 . . . 32
Problem 5.8 . . . 32
Problem 5.9 . . . 33
Problem 5.10 . . . 33

**Problem Set 6** **34**
Problem 6.1 . . . 34
Problem 6.2 . . . 34
Problem 6.3 . . . 34
Problem 6.4 . . . 34
Problem 6.5 . . . 34
Problem 6.6 . . . 35
Problem 6.7 . . . 35
Problem 6.8 . . . 35
Problem 6.9 . . . 35

**Problem Set 7 (Munkres §22)** **36**
Problem 7.1 . . . 36
Problem 7.2 . . . 36
Problem 7.3 . . . 36
Problem 7.4 . . . 36
Problem 7.5 . . . 37
Problem 7.6 . . . 37

**Problem Set 8 (Munkres §51, §52)** **38**
Problem 8.1 . . . 38
Problem 8.2 . . . 38
Problem 8.3 . . . 38
Problem 8.4 . . . 38
Problem 8.5 . . . 39
Problem 8.6 . . . 39
Problem 8.7 . . . 39
Problem 8.8 . . . 39
Problem 8.9 . . . 39
Problem 8.10 . . . 40

# Problem Set 1

## Problem 1.1

Let $A$ be a set and $\mathcal{P}(A)$ be its power set. Prove that there is no injective map $f\colon \mathcal{P}(A) \to A$ and no surjective map $g\colon A \to \mathcal{P}(A)$.

**Solution pending**

## Problem 1.2

Let $X$ be an infinite set. Is the collection $\mathcal{T}_\infty := \{U \subseteq X \mid X \setminus U \text{ is infinite or } U = \emptyset\}$ a topology on $X$? Justify your answer.

**Solution pending**

## Problem 1.3

Let $X = \{a, b, c\}$. Determine which of the following collections of subsets form a valid topology on $X$. For those that fail, explicitly state which axiom or axioms are violated:

(i) $\mathcal{T}_1 = \{\emptyset, \{a\}, \{b\}, X\}$;

(ii) $\mathcal{T}_2 = \{\emptyset, \{a\}, \{a, b\}, X\}$;

(iii) $\mathcal{T}_3 = \{\emptyset, \{a, b\}, \{b, c\}, X\}$.

**Solution pending**

## Problem 1.4

Let $X$ be a topological space and $A \subset X$. Prove that $A$ is open in $X$ if and only if for each $x \in A$, $\exists$ an open set $U$ in $X$ such that $x \in U \subseteq A$.

**Solution pending**

## Problem 1.5

Let $U$ be an open set and $V$ be a closed set in $X$. Prove that $U \setminus V$ is open in $X$ and $V \setminus U$ is closed in $X$.

**Solution pending**

## Problem 1.6

Let $X = \mathbb{N}$ and define $\mathcal{T}_1, \mathcal{T}_2 \subseteq \mathcal{P}(\mathbb{N})$ as follows:

(i) $\mathcal{T}_1 := \{\emptyset, \mathbb{N}, \{1\}, \{1,2\}, \{1,2,3\}, \{1,2,3,4\}, \dots\}$.

(ii) $\mathcal{T}_2 := \{\emptyset, \mathbb{N}, \mathbb{N} \setminus \{1\}, \mathbb{N} \setminus \{1,2\}, \mathbb{N} \setminus \{1,2,3\}, \mathbb{N} \setminus \{1,2,3,4\}, \dots\}$.

Determine whether $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

### Solution

*submitted by* **Sahil**

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Throughout, a topology on a set $X$ is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ such that (1) $\varnothing, X \in \mathcal{T}$, (2) arbitrary unions of members of $\mathcal{T}$ belong to $\mathcal{T}$, and (3) finite intersections of members of $\mathcal{T}$ belong to $\mathcal{T}$.

Let

$$A_n = \{1, 2, \dots, n\}, \qquad B_n = \mathbb{N} \setminus A_n \quad (n \geq 1).$$

Then the two collections in the question can be written as

$$\mathcal{T}_1 = \{\varnothing, \mathbb{N}\} \cup \{A_n : n \geq 1\}, \qquad \mathcal{T}_2 = \{\varnothing, \mathbb{N}\} \cup \{B_n : n \geq 1\}.$$

We verify the topology axioms separately.

**(a) The collection $\mathcal{T}_1$ is a topology.** Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_1$. For finite intersections, observe that

$$A_m \cap A_n = A_{\min\{m,n\}}.$$

Intersections with $\varnothing$ or $\mathbb{N}$ also belong to $\mathcal{T}_1$. Thus $\mathcal{T}_1$ is closed under finite intersections.

Now let $\{U_i\}_{i \in I}$ be any family of members of $\mathcal{T}_1$. If every $U_i$ is empty, then $\bigcup_{i \in I} U_i = \varnothing$. If at least one $U_i = \mathbb{N}$, the union is $\mathbb{N}$. In the remaining case, the nonempty sets in the family are of the form $A_n$. If their indices are unbounded in $\mathbb{N}$, then every positive integer belongs to some $A_n$, so their union is $\mathbb{N}$. If their indices are bounded, they have a largest value $m$ (a nonempty bounded subset of $\mathbb{N}$ has a maximum), and the union is $A_m$. In every case the union belongs to $\mathcal{T}_1$. Hence $\mathcal{T}_1$ is a topology on $\mathbb{N}$.

**(b) The collection $\mathcal{T}_2$ is a topology.** Clearly, $\varnothing, \mathbb{N} \in \mathcal{T}_2$. For finite intersections, De Morgan's law gives

$$B_m \cap B_n = (\mathbb{N} \setminus A_m) \cap (\mathbb{N} \setminus A_n) = \mathbb{N} \setminus (A_m \cup A_n) = B_{\max\{m,n\}}.$$

Intersections involving $\varnothing$ or $\mathbb{N}$ also remain in $\mathcal{T}_2$.

Consider an arbitrary family $\{U_i\}_{i \in I} \subseteq \mathcal{T}_2$. If one member is $\mathbb{N}$, its union is $\mathbb{N}$. If every member is empty, its union is empty. Otherwise, discard the empty members and write the remaining members as $B_{n_i}$. The set of indices $\{n_i : i \in I,\ U_i \neq \varnothing\}$ is a nonempty subset of $\mathbb{N}$, so it has a least element, say $m$. Since the sets $B_n$ increase as $n$ decreases, we have

$$\bigcup_{i \in I} B_{n_i} = B_m.$$

Indeed, $B_m$ is one of the sets in the union and contains every $B_{n_i}$ because $m \leq n_i$. Thus the union belongs to $\mathcal{T}_2$. Therefore $\mathcal{T}_2$ is also a topology on $\mathbb{N}$.

**Conclusion:** Both $\mathcal{T}_1$ and $\mathcal{T}_2$ define topologies on $\mathbb{N}$.

## Problem 1.7

Show that if $\mathscr{B}$ is a basis for a topology on $X$, then the topology generated by $\mathscr{B}$ equals the intersection of all topologies on $X$ that contain $\mathscr{B}$.

### Solution

*submitted by* **Sahil**

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Let $\mathcal{B}$ be a basis for a topology on $X$, and let $\mathcal{T}_{\mathcal{B}}$ denote the topology generated by $\mathcal{B}$. By the definition of the topology generated by a basis,

$$\mathcal{T}_{\mathcal{B}} = \Big\{ \bigcup_{\alpha \in A} B_\alpha : A \text{ is an index set and } B_\alpha \in \mathcal{B} \Big\}.$$

Let $\mathfrak{F}$ be the collection of all topologies on $X$ that contain $\mathcal{B}$. We prove

$$\boxed{\mathcal{T}_{\mathcal{B}} = \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.}$$

First, let $U \in \mathcal{T}_{\mathcal{B}}$. Then $U$ is a union of elements of $\mathcal{B}$. If $\mathcal{T} \in \mathfrak{F}$, then $\mathcal{B} \subseteq \mathcal{T}$. Since every topology is closed under arbitrary unions, it follows that $U \in \mathcal{T}$. This holds for every $\mathcal{T} \in \mathfrak{F}$, and hence

$$\mathcal{T}_{\mathcal{B}} \subseteq \bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T}.$$

For the reverse inclusion, note that $\mathcal{T}_{\mathcal{B}}$ is itself a topology on $X$ containing $\mathcal{B}$. Consequently, $\mathcal{T}_{\mathcal{B}} \in \mathfrak{F}$. The intersection of all members of $\mathfrak{F}$ is therefore contained in this particular member:

$$\bigcap_{\mathcal{T} \in \mathfrak{F}} \mathcal{T} \subseteq \mathcal{T}_{\mathcal{B}}.$$

Combining the two inclusions proves the required equality.

**Interpretation:** The topology generated by a basis is the smallest topology on $X$ that contains every basis element.

## Problem 1.8

Let $X$ be an infinite set and $\mathcal{T}_f$ be the finite complement topology on $X$. If $U$ and $V$ are non-empty open sets in $(X, \mathcal{T}_f)$, prove that $U \cap V \neq \emptyset$.

### Solution

*submitted by* **Sahil**

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Let $X$ be infinite and let

$$\mathcal{T}_f = \{\varnothing\} \cup \{U \subseteq X : X \setminus U \text{ is finite}\}$$

be the finite-complement topology. Suppose $U, V \in \mathcal{T}_f$ are nonempty. By the definition of $\mathcal{T}_f$, both complements $X \setminus U$ and $X \setminus V$ are finite. Using De Morgan's law,

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V).$$

The right-hand side is a union of two finite sets, and is therefore finite. If $U \cap V = \varnothing$, then the left-hand side equals $X$, which would imply that $X$ is finite. This contradicts the assumption that $X$ is infinite. Hence

$$\boxed{U \cap V \neq \varnothing.}$$

## Problem 1.9

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *particular point* collection

$$\mathcal{T}_a = \{V \subseteq X \mid V = \emptyset \text{ or } a \in V\}$$

defines a valid topology on $X$.

### Solution

*submitted by* **Sahil**

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Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a = \{V \subseteq X : V = \varnothing \text{ or } a \in V\}.$$

We check the topology axioms.

**(i) Empty set and whole space.** By definition, $\varnothing \in \mathcal{T}_a$. Also $a \in X$, so $X \in \mathcal{T}_a$.

**(ii) Arbitrary unions.** Let $\{V_i\}_{i \in I}$ be any family in $\mathcal{T}_a$. If every $V_i$ is empty, then $\bigcup_{i \in I} V_i = \varnothing$. Otherwise, at least one member, say $V_j$, is nonempty. Since $V_j \in \mathcal{T}_a$, it contains $a$. Therefore $a \in \bigcup_{i \in I} V_i$, so the union belongs to $\mathcal{T}_a$.

**(iii) Finite intersections.** It is enough to check intersections of two members, since repeated application gives the result for every finite intersection. Let $U, V \in \mathcal{T}_a$. If either set is empty, then $U \cap V = \varnothing \in \mathcal{T}_a$. If neither is empty, both contain $a$, and hence $a \in U \cap V$. Thus $U \cap V \in \mathcal{T}_a$.

All three axioms hold. Therefore $\boxed{\mathcal{T}_a \text{ is a topology on } X.}$

## Problem 1.10

Let $X \neq \emptyset$ be a set and let $a \in X$ be a fixed point. Prove that the *excluded point* collection

$$\mathcal{T}^a := \{V \subseteq X \mid V = X \text{ or } a \notin V\}$$

defines a valid topology on $X$.

### Solution

*submitted by* **Sahil**

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Let $X \neq \varnothing$ and fix $a \in X$. Define

$$\mathcal{T}_a^e = \{V \subseteq X : V = X \text{ or } a \notin V\}.$$

Thus the open sets are $X$ itself and all subsets of $X$ that do not contain $a$. We verify the topology axioms.

**(i) Empty set and whole space.** Since $a \notin \varnothing$, we have $\varnothing \in \mathcal{T}_a^e$. Also $X \in \mathcal{T}_a^e$ by definition.

**(ii) Arbitrary unions.** Let $\{V_i\}_{i\in I}$ be any family in $\mathcal{T}_a^e$. If some $V_j = X$, then $\bigcup_{i\in I} V_i = X$. If no member equals $X$, then every $V_i$ omits $a$. Consequently, their union also omits $a$:

$$a \notin V_i \text{ for all } i \in I \quad \Longrightarrow \quad a \notin \bigcup_{i\in I} V_i.$$

In either case the union belongs to $\mathcal{T}_a^e$.

**(iii) Finite intersections.** Again it is enough to consider two sets. Let $U, V \in \mathcal{T}_a^e$. If both equal $X$, then $U \cap V = X$. Otherwise, at least one of them, say $U$, is not $X$. By the definition of the collection, $a \notin U$. Since $U \cap V \subseteq U$, the intersection also omits $a$. Hence $U \cap V \in \mathcal{T}_a^e$.

All three topology axioms are satisfied. Therefore

$$\boxed{\mathcal{T}_a^e \text{ is a topology on } X.}$$

## Problem 1.11

Let $X$ be an infinite set and let $a \in X$ be a fixed point. Define a collection of subsets $\mathcal{T}_a \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T}_a = \mathcal{P}(X \setminus \{a\}) \cup \{V \subseteq X \mid a \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Prove that $\mathcal{T}_a$ defines a valid topology on $X$. What happens when $X$ is a finite set?

**Solution pending**

## Problem 1.12

Let $X$ be an infinite set and let $a, b \in X$ be two distinct points. Define a collection of subsets $\mathcal{T} \subseteq \mathcal{P}(X)$ by:

$$\mathcal{T} = \mathcal{P}(X \setminus \{a, b\}) \cup \{V \subseteq X \mid a \in V \text{ or } b \in V \text{ and } X \setminus V \text{ is finite}\}.$$

Is $\mathcal{T}$ a topology on $X$?

**Solution pending**

## Problem 1.13

Let $\mathscr{B}_1$, $\mathscr{B}_2$, and $\mathscr{B}_3$ be respectively the collections of all open balls, open rectangles, and open triangles in the plane $\mathbb{R}^2$.

(i). Prove that all three collections satisfy the basis axioms on $\mathbb{R}^2$.

(ii). Let $\mathcal{T}_i$ be the topology generated by the basis $\mathscr{B}_i$ for $i = 1, 2, 3$. Let $f\colon (\mathbb{R}^2, \mathcal{T}_i) \to (\mathbb{R}^2, \mathcal{T}_j)$

be the identity map. Prove that for any choice of $i, j \in \{1, 2, 3\}$, the map $f$ is continuous.

(iii). Conclude that $\mathcal{T}_1 = \mathcal{T}_2 = \mathcal{T}_3$.

**Solution pending**

## Problem 1.14

Along with the finite complement topology ($\mathcal{T}_f$) and the countable complement topology ($\mathcal{T}_c$), we consider the topologies generated by the following six bases:

(i) **Standard Topology ($\mathcal{T}_{\text{std}}$):** Generated by the basis of all open intervals:

$$\mathscr{B}_1 = \{(a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(ii) **$K$-Topology ($\mathcal{T}_K$):** Generated by the basis:

$$\mathscr{B}_2 = \{(a, b) \mid a < b\} \cup \{(a, b) \setminus K \mid a < b\}, \quad \text{where } K = \{1, \tfrac{1}{2}, \tfrac{1}{3}, \dots\}.$$

(iii) **Lower Limit Topology ($\mathcal{T}_\ell$):** Generated by the basis of all half-open intervals closed on the left:

$$\mathscr{B}_3 = \{[a, b) \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(iv) **Upper Limit Topology ($\mathcal{T}_u$):** Generated by the basis of all half-open intervals closed on the right:

$$\mathscr{B}_4 = \{(a, b] \mid a < b \text{ and } a, b \in \mathbb{R}\}.$$

(v) **Left-Ray Topology ($\mathcal{T}_{\text{left}}$):** Generated by the basis of open left rays:

$$\mathscr{B}_5 = \{(-\infty, b) \mid b \in \mathbb{R}\}.$$

(vi) **Right-Ray Topology ($\mathcal{T}_{\text{right}}$):** Generated by the basis of open right rays:

$$\mathscr{B}_6 = \{(a, \infty) \mid a \in \mathbb{R}\}.$$

For each pair of these topologies, determine whether one contains the other (i.e., whether one is finer, coarser, or if they are completely incomparable). Summarize your results by drawing a partial order hierarchy diagram representing the containment relations among all eight topologies.

**Solution pending**

## Problem 1.15

Let $\mathcal{T}_X$ denote the collection of all topologies on a non-empty set $X$ and let $\mathcal{T}_X/\text{hom}$ be the set of homeomorphism classes of topologies on $X$. Let $S_X$ be the group of all bijective maps on $X$. Note that each bijection $f\colon X \to X$ induces a bijection $\tilde{f}\colon \mathcal{P}(X) \to \mathcal{P}(X)$ defined by $\tilde{f}(A) = \{f(a) \mid a \in A\}$ for every subset $A \in \mathcal{P}(X)$. Consequently, $S_X$ acts naturally on $\mathcal{T}_X$

via $f.\mathcal{T} = \tilde{f}(\mathcal{T})$ for all $f \in S_X$ and $\mathcal{T} \in \mathcal{T}_X$. Prove that there is a natural bijection between $\mathcal{T}_X/\text{hom}$ and the collection of orbits of the natural action of $S_X$ on $\mathcal{T}_X$.

**Solution pending**

### Problem 1.16

List down all possible (there are 29) topologies on a three-point set $X = \{a, b, c\}$. Compute the number of orbits under the natural action of $S_3$ on the set $\mathcal{T}_X$ of all topologies on $X$.

**Solution pending**

### Problem 1.17

Let $X$ be a set, and let $\mathcal{S}_1$ and $\mathcal{S}_2$ be two subbases on $X$ generating topologies $\mathcal{T}_1$ and $\mathcal{T}_2$, respectively.

(i). If $\mathcal{S}_1 \subseteq \mathcal{S}_2$, prove that $\mathcal{T}_1 \subseteq \mathcal{T}_2$.

(ii). Give an explicit example where $\mathcal{T}_1 = \mathcal{T}_2$ even though $\mathcal{S}_1 \neq \mathcal{S}_2$.

**Solution pending**

# Problem Set 2

## Problem 2.1

Whether each of the following subsets of $\mathbb{R}^2$ can be written as a Cartesian product $A \times B$, where $A, B \subset \mathbb{R}$?

(i) $\{(x, y) \mid x \text{ is an integer}\}$.

(ii) $\{(x, y) \mid 0 \leq x < 1\}$.

(iii) $\{(x, y) \mid x < y\}$.

(iv) $\{(x, y) \mid x \text{ is an integer, and } y \text{ is not a rational number}\}$.

(v) $\{(x, y) \mid x^2 + y^2 \leq 2026\}$.

**Solution pending**

## Problem 2.2

Let $f \colon A \to B$ be a function, with $A_0 \subseteq A$ and $B_0 \subseteq B$.

(i) Show that $A_0 \subseteq f^{-1}(f(A_0))$, and that equality holds for every subset $A_0 \subseteq A$ if and only if $f$ is injective.

(ii) Show that $f(f^{-1}(B_0)) \subseteq B_0$, and that equality holds for every subset $B_0 \subseteq B$ if and only if $f$ is surjective.

**Solution pending**

## Problem 2.3

Let $f \colon A \to B$ be a function, and $B_0, B_1 \subseteq B$. Show that preimages preserve inclusions, unions, intersections, and set differences by proving the following:

(i) If $B_0 \subseteq B_1$, then $f^{-1}(B_0) \subseteq f^{-1}(B_1)$.

(ii) $f^{-1}(B_0 \cup B_1) = f^{-1}(B_0) \cup f^{-1}(B_1)$.

(iii) $f^{-1}(B_0 \cap B_1) = f^{-1}(B_0) \cap f^{-1}(B_1)$.

(iv) $f^{-1}(B_0 \setminus B_1) = f^{-1}(B_0) \setminus f^{-1}(B_1)$.

**Solution pending**

## Problem 2.4

Let $f\colon A \to B$ be a function, and let $A_0, A_1 \subseteq A$.

(i) Show that images preserve inclusions and unions:

(a) If $A_0 \subseteq A_1$, then $f(A_0) \subseteq f(A_1)$.

(b) $f(A_0 \cup A_1) = f(A_0) \cup f(A_1)$.

(ii) Show that images satisfy only one-way inclusions for intersections and set differences in general:

(a) $f(A_0 \cap A_1) \subseteq f(A_0) \cap f(A_1)$.

(b) $f(A_0) \setminus f(A_1) \subseteq f(A_0 \setminus A_1)$.

(iii) Prove that equality holds in both (ii)(a) and (ii)(b) for all subsets $A_0, A_1 \subseteq A$ if and only if $f$ is injective.

**Solution pending**

## Problem 2.5

Let $\sim$ be a relation defined on the Euclidean plane $\mathbb{R}^2$ as follows: two points are related if and only if their distances from the origin are equal.

(i) Show that $\sim$ is an equivalence relation (i.e., reflexivity, symmetry, and transitivity).

(ii) Describe the equivalence class $[P]$ for any point $P \in \mathbb{R}^2$.

(iii) What is the geometric interpretation of the partition of $\mathbb{R}^2$ induced by this relation?

**Solution pending**

## Problem 2.6

Let $X$ be a non-empty set. Suppose that $S$ is the collection of the following two types of open rays: $(a, \infty) = \{x \in X \mid a < x\}$, and $(-\infty, b) = \{x \in X \mid x < b\}$. Prove that $S$ is a subbasis, and it generates the order topology on $X$.

**Solution pending**

## Problem 2.7

Let $\{A_n\}_{n=1}^{\infty}$ be a sequence of non-empty countable sets.

(1) Prove that the finite Cartesian product $\prod_{i=1}^{k} A_i = A_1 \times A_2 \times \cdots \times A_k$ is countable for any $k \in \mathbb{N}$.

(2) Determine whether the infinite Cartesian product $\prod_{n=1}^{\infty} A_n$ is necessarily countable. Jus-

tify your answer.

**Solution pending**

### Problem 2.8

Let $X$ be a linearly ordered set. Prove that a singleton set $\{x\}$ is open in the order topology on $X$ if and only if $x$ has an immediate predecessor and an immediate successor, or if $x$ is the minimum/maximum and has an immediate successor/predecessor respectively.

**Solution pending**

### Problem 2.9

Prove that if $V$ is closed in $Y$ and $Y$ is closed in $X$, then $V$ is closed in $X$.

**Solution pending**

### Problem 2.10

Let $Y$ be a subspace of $X$ and let $A$ be a subset of $Y$. Prove that the topology $A$ inherits as a subspace of $Y$ is identical to the topology it inherits directly as a subspace of $X$.

**Solution pending**

### Problem 2.11

Let $A$ be a subset of a topological space $X$ and $x \in X$ be an element. Then prove that the following statements are equivalent:

(i). The point $x \in \overline{A}$.

(ii). Every neighborhood of $x$ intersects $A$.

(iii). Every basis element containing $x$ intersects $A$.

**Solution pending**

### Problem 2.12

Prove that $\overline{A} = A \cup A'$. Deduce that $A$ is closed in $X$ if and only if $A' \subseteq A$.

**Solution pending**

### Problem 2.13

Prove that $\operatorname{Int} A \cap \operatorname{Bd} A = \emptyset$, and $\overline{A} = \operatorname{Int} A \cup \operatorname{Bd} A$.

**Solution pending**

### Problem 2.14

Prove that every order topology is Hausdorff.

**Solution pending**

### Problem 2.15

Let $X$ be a $T_1$ space. Then prove that for each pair $x_1, x_2$ of distinct points, there exist open sets $U_1$ and $U_2$ containing $x_1$ and $x_2$ respectively such that $x_1 \notin U_2$ and $x_2 \notin U_1$.

**Solution pending**

# Problem Set 3

## Problem 3.1

Give an example of a $T_1$ space where a sequence converges to more than one point. Conclude that a $T_1$ space is not necessarily a Hausdorff space.

**Solution** *submitted by* **Arunodaya**

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Consider the set $X = \mathbb{R}$ equipped with the cofinite topology

$$\tau = \{\emptyset\} \cup \{U \subseteq \mathbb{R} : \mathbb{R} \setminus U \text{ is finite}\},$$

that is, the open sets are the empty set or subsets of $\mathbb{R}$ whose complements are finite.

Now, we know that a topological space is $T_1$ if every singleton set is closed. Let $x \in \mathbb{R}$. Then $\mathbb{R} \setminus \{x\}$ is open because its complement $\{x\}$ is finite. Therefore, $\{x\}$ is closed. Since $x$ is arbitrary, every singleton subset of $\mathbb{R}$ is closed. Hence $(\mathbb{R}, \tau)$ is a $T_1$ space.

Consider the sequence

$$x_n = n, \quad n \in \mathbb{N}.$$

$x_n \to x$ if for every open neighbourhood $U$ of $x$, $\exists\, N \in \mathbb{N}$ such that $n \geq N \Rightarrow x_n \in U$.

Let $x \in \mathbb{R}$ be arbitrary. Let $U$ be any open neighbourhood of $x$. By the definition of the cofinite topology, $\mathbb{R} \setminus U$ is finite. Since $x_n = n$, the sequence consists of distinct natural numbers. Consequently only finitely many terms of the sequence can belong to the finite set $\mathbb{R} \setminus U$. Therefore, $\exists\, N \in \mathbb{N}$ such that

$$n \geq N \Rightarrow n \notin \mathbb{R} \setminus U.$$

Thus, $n \geq N \Rightarrow x_n = n \in U$. So, by the definition of convergence,

$$x_n \to x \quad \forall\, x \in \mathbb{R}.$$

Now, the sequence $x_n = n$ converges to every real number

$\Rightarrow$ its limit is not unique

$\Rightarrow$ the space cannot be Hausdorff.

Thus $(\mathbb{R}, \tau)$ is $T_1$ but not $T_2$, i.e. $T_1 \not\Rightarrow T_2$.

## Problem 3.2

Prove that the product topology $\mathcal{T}_{\text{prod}}$ on $\mathbb{R}^n$ (where each factor $\mathbb{R}$ is equipped with its standard Euclidean topology) coincides with the Euclidean topology $\mathcal{T}_{\text{Euc}}$ on $\mathbb{R}^n$ induced by the standard Euclidean metric.

**Solution** *submitted by* **Arunodaya**

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Let $U = U_1 \times U_2 \times \cdots \times U_n$ be a basic open set in the product topology. Take an arbitrary point $x = (x_1, x_2, x_3, x_4, \ldots, x_n) \in U$. Since each $U_i$ is open in $\mathbb{R}$, $\exists\, \varepsilon_i > 0$ such that

$$(x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i.$$

Choose $\varepsilon = \min\{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n\}$. Since there are finitely many positive numbers, $\varepsilon > 0$.

Now, consider any point $y = (y_1, y_2, \ldots, y_n) \in B(x, \varepsilon)$. Then $d(x, y) < \varepsilon$. Thus,

$$\sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \varepsilon \;\Rightarrow\; \sum_{i=1}^{n}(x_i - y_i)^2 < \varepsilon^2.$$

Since every term in the sum is non-negative, $(x_i - y_i)^2 < \varepsilon^2$ for every $i = 1, 2, \ldots, n$. Consequently $|x_i - y_i| < \varepsilon \leq \varepsilon_i$; thus $y_i \in (x_i - \varepsilon_i,\, x_i + \varepsilon_i) \subseteq U_i$. This holds for every $i$. Hence

$$y \in U_1 \times U_2 \times \cdots \times U_n \;\Rightarrow\; B(x, \varepsilon) \subseteq U.$$

Since $x \in U$ was arbitrary, every point of $U$ has a Euclidean open ball contained in $U$. Hence $U$ is Euclidean open.

$$\underline{\mathcal{T}_{\text{prod}} \subseteq \mathcal{T}_{\text{Euc}}}$$

Let $V$ be an arbitrary Euclidean open subset of $\mathbb{R}^n$. Take any point $x = (x_1, x_2, \ldots, x_n) \in V$. Since $V$ is Euclidean open, $\exists\, \delta > 0$ such that $B(x, \delta) \subseteq V$. Now consider the product of open intervals

$$W = \prod_{i=1}^{n}\Big(x_i - \frac{\delta}{\sqrt{n}},\; x_i + \frac{\delta}{\sqrt{n}}\Big).$$

We will show that $W \subseteq B(x, \delta)$. Take any $y = (y_1, y_2, y_3, y_4, \ldots, y_n) \in W$. By definition of $W$,

$$|x_i - y_i| < \delta/\sqrt{n} \quad \forall\, i = 1, 2, 3, 4, \ldots, n.$$

Therefore

$$d(x, y) = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2} < \sqrt{\sum_{i=1}^{n}\frac{\delta^2}{n}} = \sqrt{\frac{n\delta^2}{n}} = \delta.$$

Thus $y \in B(x, \delta)$. Consequently $W \subseteq B(x, \delta) \subseteq V$.

Notice that $W$ is a basic open set in the product topology because it is a product of open intervals in $\mathbb{R}$. Hence, every point $x \in V$ has a basic product open neighbourhood contained in $V$. Therefore, $V$ is open in the product topology. So, $\underline{\mathcal{T}_{\text{Euc}} \subseteq \mathcal{T}_{\text{prod}}}$.

So $\mathcal{T}_{\text{prod}} = \mathcal{T}_{\text{Euc}}$.

## Problem 3.3

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and let $\phi\colon \Lambda \to \Lambda$ be a bijection. Prove that the product spaces $X = \prod_{\alpha\in\Lambda} X_\alpha$ and $X_\phi = \prod_{\alpha\in\Lambda} X_{\phi(\alpha)}$, both equipped with their respective product topologies, are homeomorphic.

### Solution

*submitted by* **Arunodaya**

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We define $f\colon X \to X_\phi$ by

$$f\big((x_\alpha)_{\alpha\in A}\big) = \big(x_{\phi(\alpha)}\big)_{\alpha\in A}.$$

Since $\phi$ is a bijection, every coordinate in the second product corresponds to exactly one coordinate in the first product. Since $\phi$ is bijective, it has an inverse

$$\phi^{-1}\colon B \to A.$$

Define $g\colon X_\phi \to X$ by

$$g\big((y_\beta)_{\beta\in B}\big) = \big(y_{\phi^{-1}(\beta)}\big)_{\beta\in B}.$$

Then for every $x = (x_\alpha) \in X$, $g(f(x)) = x$. Similarly for every $y \in X_\phi$, $f(g(y)) = y$. Thus $g = f^{-1}$. Hence $f$ is bijective.

The product topology has a subbasis consisting of sets of the form $\pi_\alpha^{-1}(U_\alpha)$, where $U_\alpha$ is open in $X_\alpha$ and $\pi_\alpha\colon X \to X_\alpha$ is the coordinate projection.

Let $U_\alpha$ be open in $X_{\phi(\alpha)}$. Consider the subbasic open set in $X_\phi$

$$V = \{y \in X_\phi : y_\alpha \in U_\alpha\}.$$

$$\begin{aligned} f^{-1}(V) &= \{x \in X : f(x)_\alpha \in U_\alpha\} \\ &= \{x \in X : x_\alpha \in U_\alpha\} \\ &= \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

Since $U_\alpha$ is open and $\pi_\alpha$ is continuous, the set is open in $X \Rightarrow$ the inverse image of every subbasic open set is open. Consequently, $f$ is continuous.

By the same argument, for every subbasic open set

$$W = \{x \in X : x_\alpha \in U_\alpha\}$$

we have

$$(f^{-1})^{-1}(W) = \{y \in X_\phi : y_{\phi^{-1}(\alpha)} \in U_\alpha\};$$

this is a subbasic open set in $X_\phi$. Hence $f^{-1}$ is continuous.

We proved that $f$ is bijective, continuous and has a continuous inverse; therefore $X \cong X_\phi$.

## Problem 3.4

Let $\{X_\alpha\}_{\alpha\in\Lambda}$ be a family of topological spaces, and for each $\alpha \in \Lambda$, let $Y_\alpha \subseteq X_\alpha$ be equipped with the subspace topology. Prove that the product topology on $Y = \prod_{\alpha\in\Lambda} Y_\alpha$ coincides with the subspace topology on $Y$ induced by the product topology on $X = \prod_{\alpha\in\Lambda} X_\alpha$.

### Solution

*submitted by* **Arunodaya**

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Let

- $\tau_1$ denote the product topology on $Y$,
- $\tau_2$ denote the subspace topology on $Y$ inherited from $X$.

We shall prove $\tau_1 = \tau_2$.

Let $B$ be a basic open set in $X$. Then

$$B = \bigcap_{\alpha\in F} \pi_\alpha^{-1}(U_\alpha),$$

where each $U_\alpha$ is open in $X_\alpha$. By the definition of the subspace topology, a basic open set in $Y$ is $B \cap Y$. Now,

$$\begin{aligned} B \cap Y &= Y \cap \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha) \\ &= \bigcap_{\alpha \in F} \left(Y \cap \pi_\alpha^{-1}(U_\alpha)\right) \\ &= \bigcap_{\alpha \in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha). \end{aligned}$$

Since $U_\alpha$ is open in $X_\alpha$, the set $U_\alpha \cap Y_\alpha$ is open in $Y_\alpha$ by the subspace topology. Therefore, $B \cap Y$ is open in the product topology on $Y$. Hence $\underline{\tau_2 \subseteq \tau_1}$.

Let $C$ be a basic open set in the product topology on $Y$. Then

$$C = \bigcap_{\alpha \in F} q_\alpha^{-1}(V_\alpha),$$

where $V_\alpha$ is open in $Y_\alpha$. Since $Y_\alpha$ has the subspace topology inherited from $X_\alpha$, $\exists$ an open set $U_\alpha$ in $X_\alpha$ such that $V_\alpha = U_\alpha \cap Y_\alpha$. Therefore,

$$\begin{aligned} C &= \bigcap_{\alpha \in F} q_\alpha^{-1}(U_\alpha \cap Y_\alpha) \\ &= Y \cap \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha). \end{aligned}$$

But $\bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$ is open in $X$ because it is a basic open set in the product topology on $X$. Thus, $C$ is the intersection of $Y$ with an open set in $X$. Hence $C$ is open in the subspace topology on $Y$. Therefore $\underline{\tau_1 \subseteq \tau_2}$.

So, $\underline{\tau_1 = \tau_2}$.

## Problem 3.5

Let $\{X_\alpha\}_{\alpha \in \Lambda}$ be an indexed collection of topological spaces, and let $X = \prod_{\alpha \in \Lambda} X_\alpha$ be equipped with the product topology $\mathcal{T}_{\text{prod}}$. For any fixed index $\beta \in \Lambda$, consider the $\beta$-th coordinate projection map

$$p_\beta \colon \prod_{\alpha \in \Lambda} X_\alpha \longrightarrow X_\beta, \qquad p_\beta\big((x_\alpha)_{\alpha \in \Lambda}\big) = x_\beta.$$

Prove that $p_\beta$ is an open map. Is $p_\beta$ necessarily a closed map?

### Solution

*submitted by* **Arunodaya**

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**Part (i).** By the definition of the product topology, every basic open set has the form

$$B = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha),$$

where

- $F \subseteq A$ is finite,
- each $U_\alpha$ is open in $X_\alpha$.

**Case 1:** $\beta \in F$. Then $B = \bigcap_{\alpha \in F} \pi_\alpha^{-1}(U_\alpha)$, and the image of $B$ under $p_\beta$ is

$$p_\beta(B) = U_\beta,$$

provided the other coordinate restrictions are non-empty. Indeed, every element of $B$ has its $\beta$-coordinate in $U_\beta$. Conversely, any element of $U_\beta$ can be extended to a point of $B$ if all the other coordinate restrictions are non-empty. Since $U_\beta$ is open in $X_\beta$, the image is open. If another coordinate restriction is empty, then $B = \emptyset$, whose image is also open.

**Case 2:** $\beta \notin F$. In this case, there is no restriction on the $\beta$-coordinate:

$$p_\beta(B) = X_\beta,$$

provided $B \neq \emptyset$. Since $X_\beta$ is open in itself, the image is open. If $B = \emptyset$, its image is open again.

Now, $U = \bigcup_{\lambda \in \Lambda} B_\lambda$. Using the fact that images preserve arbitrary unions, we get

$$p_\beta(U) = p_\beta\Big(\bigcup_{\lambda \in \Lambda} B_\lambda\Big) = \bigcup_{\lambda \in \Lambda} p_\beta(B_\lambda).$$

From the above step, $p_\beta(B_\lambda)$ is open in $X_\beta$. An arbitrary union of open sets is open; thus, $p_\beta(U)$ is open in $X_\beta$.

**Part (ii). No.** Take $X_1 = \mathbb{R}$, $X_2 = \mathbb{R}$ with their usual Euclidean topologies. Then $X = X_1 \times X_2 = \mathbb{R}^2$. Consider the coordinate projection $p_1 \colon \mathbb{R}^2 \to \mathbb{R}$ defined by $p_1(x, y) = x$. Consider

$$F = \{(x, y) \in \mathbb{R}^2 : xy = 1\}.$$

Define $h \colon \mathbb{R}^2 \to \mathbb{R}$ by $h(x, y) = xy$. Since multiplication is continuous, $h$ is continuous. Also $F = h^{-1}(\{1\})$. Since $\{1\}$ is closed in $\mathbb{R}$ and the inverse image of a closed set under a continuous function is closed, we say $F$ is closed in $\mathbb{R}^2$.

We have

$$p_1(F) = \{x \in \mathbb{R} \mid \exists\, y \in \mathbb{R} \text{ such that } xy = 1\} = \mathbb{R} \setminus \{0\}.$$

Indeed, if $x \neq 0$ we can choose $y = 1/x$. But if $x = 0$, no such $y$ exists. Therefore $p_1(F) = \mathbb{R} \setminus \{0\}$.

The set $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is not closed in $\mathbb{R}$ because its complement $\{0\}$ is not open in $\mathbb{R}$. Thus, we found a closed set $F \subseteq \mathbb{R}^2$ whose image under $p_1$ is not closed. Consequently, $p_1$ is not closed.

So, every coordinate projection is an open map. A coordinate projection need not be a closed map.

## Problem 3.6

Let $L$ be a line in the plane $\mathbb{R}^2$, and $\mathbb{R}_\ell$ be the lower limit topology on $\mathbb{R}$. Discuss the topology $L$ inherits as a subspace of $\mathbb{R}_\ell \times \mathbb{R}$ and as a subspace of $\mathbb{R}_\ell \times \mathbb{R}_\ell$.

### Solution

*submitted by* **Pula Karan**

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We analyze the subspace topology on $L$ under both product topologies.

**Part 1: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}$.** A basis for $\mathbb{R}_l \times \mathbb{R}$ consists of rectangles of the form $[a, b) \times (c, d)$ with $a < b$ and $c < d$.

(a) **Vertical Line ($L : x = k$):** For any $(k, y_0) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [k, k + \epsilon) \times (y_0 - \epsilon, y_0 + \epsilon)$. Intersecting $U$ with $L$ yields:

$$U \cap L = \{k\} \times (y_0 - \epsilon, y_0 + \epsilon)$$

This traces open intervals along the $y$-axis. Thus, $L$ inherits the **standard Euclidean topology** ($\mathbb{R}$).

(b) **Horizontal Line ($L : y = k$):** For any $(x_0, k) \in L$, a basic open neighborhood in $\mathbb{R}_l \times \mathbb{R}$ is $U = [x_0, x_0 + \epsilon) \times (k - \epsilon, k + \epsilon)$. Intersecting $U$ with $L$ yields:

$$U \cap L = [x_0, x_0 + \epsilon) \times \{k\}$$

This traces right-half-open intervals along the $x$-axis. Thus, $L$ inherits the **lower limit topology** ($\mathbb{R}_l$).

(c) **Non-Vertical Line ($L : y = mx + k$, $m \neq 0$):** For any $(x_0, y_0) \in L$, consider basic open sets $U = [x_0, x_0 + \epsilon) \times (y_0 - \delta, y_0 + \delta)$.

- If $m > 0$, as $x$ increases from $x_0$, $y$ increases from $y_0$. For sufficiently small $\delta > 0$, $U \cap L$ isolates points corresponding to $x \in [x_0, x_0 + \delta/m)$.
- If $m < 0$, as $x$ increases from $x_0$, $y$ decreases from $y_0$. Intersecting $[x_0, x_0 + \epsilon)$ with $(y_0 - \delta, y_0 + \delta)$ again isolates points corresponding to $x \in [x_0, x_0 + \delta/|m|)$.

In both cases, basic open sets induce half-open intervals $[x_0, x_0 + r)$ along $L$. Thus, $L$ inherits the **lower limit topology** ($\mathbb{R}_l$).

**Part 2: Subspace Topology inherited from $\mathbb{R}_l \times \mathbb{R}_l$ (Sorgenfrey Plane).** A basis for $\mathbb{R}_l \times \mathbb{R}_l$ consists of half-open rectangles of the form $[a, b) \times [c, d)$.

(a) **Vertical ($x = k$), Horizontal ($y = k$), and Positive Slope ($y = mx + k$, $m > 0$):** For any $(x_0, y_0) \in L$, intersecting $[x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon')$ with $L$ isolates points corresponding to $[x_0, x_0 + \delta)$ where $\delta = \min(\epsilon, \epsilon'/m)$ for $m > 0$ (or $\epsilon$ if vertical/horizontal). Thus, $L$ inherits the **lower limit topology** ($\mathbb{R}_l$).

(b) **Oblique Line with Negative Slope ($y = mx + k$, $m < 0$):** Let $(x_0, y_0) \in L$ and consider $U = [x_0, x_0 + \epsilon) \times [y_0, y_0 + \epsilon)$. For any $x > x_0$, $y = mx + k < mx_0 + k = y_0$, so $(x, y) \notin U$. For $x < x_0$, $x \notin [x_0, x_0 + \epsilon)$, so $(x, y) \notin U$. Therefore, $U \cap L = \{(x_0, y_0)\}$. Since every singleton is open, $L$ inherits the **discrete topology**.

## Problem 3.7

Compare (i) the ordered square $I_o^2$, (ii) the topology $I^2$ inherits as a subspace of the product topology $\mathbb{R}^2$, and (iii) the product topology on $I \times I$.

### Solution

*submitted by* **Pula Karan**

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Let $I = [0, 1]$. We compare the three topologies on $I^2 = [0, 1] \times [0, 1]$:

1. **Topologies (ii) and (iii) are identical ($\mathcal{T}_{\text{sub}} = \mathcal{T}_{\text{prod}}$):**

- Topology (ii) is the subspace topology inherited from the Euclidean product topology on $\mathbb{R}^2$. Basis elements are of the form $(U \times V) \cap I^2$, where $U, V \subseteq \mathbb{R}$ are open intervals.
- Topology (iii) is the product topology on $I \times I$, where each factor $I$ carries the subspace topology from $\mathbb{R}$. Basis elements are of the form $(U \cap I) \times (V \cap I) = (U \times V) \cap I^2$.

Since their bases coincide, Topologies (ii) and (iii) are identical.

2. **Topology (i) ($I_o^2$) is strictly finer than Topologies (ii) and (iii):** The ordered square $I_o^2$ is equipped with the dictionary order topology $\leq_{\text{dic}}$.
   - *Euclidean $\subseteq$ Dictionary Order*: Any standard Euclidean basic open set $U = (a, b) \times (c, d) \subset (0, 1)^2$ can be written as:
   $$U = \bigcup_{x_0 \in (a,b)} \Big( \{x_0\} \times (c, d) \Big) = \bigcup_{x_0 \in (a,b)} \Big( (x_0, c), (x_0, d) \Big)_{\text{dic}}$$
   Since each vertical interval $\{x_0\} \times (c, d)$ is open in the dictionary order topology, $U$ is open in $I_o^2$.
   - *Strict Containment*: Sets of the form $\{x_0\} \times (c, d)$ for $0 < c < d < 1$ are open in $I_o^2$. However, they contain no 2D Euclidean open ball, so they are not open in the standard topology on $I^2$.

**Conclusion:** Topology (ii) = Topology (iii) $\subsetneq$ Topology (i).

## Problem 3.8

Let $X$ be a topological space, and $\Delta = \{(x, x) \in X \times X \mid x \in X\}$ be the diagonal of $X \times X$, equipped with the subspace topology inherited from the product topology on $X \times X$.

(i) Prove that $X$ is Hausdorff if and only if the diagonal $\Delta$ is a closed subset of $X \times X$.

(ii) Prove that the diagonal map $d \colon X \to \Delta$ defined by $d(x) = (x, x)$ is a homeomorphism.

### Solution

*submitted by* **Pula Karan**

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### Part (i).

*Proof.* ( $\Longrightarrow$ ) Suppose $X$ is Hausdorff. We show that $(X \times X) \setminus \Delta$ is open. Let $(x, y) \in (X \times X) \setminus \Delta$, which means $x \neq y$. Since $X$ is Hausdorff, there exist disjoint open sets $U, V \subseteq X$ such that $x \in U$, $y \in V$, and $U \cap V = \varnothing$. Then $U \times V$ is an open neighborhood of $(x, y)$ in $X \times X$. If there were some $(z, z) \in (U \times V) \cap \Delta$, then $z \in U \cap V = \varnothing$, a contradiction. Hence $(U \times V) \cap \Delta = \varnothing$, so $U \times V \subseteq (X \times X) \setminus \Delta$. Thus $(X \times X) \setminus \Delta$ is open, so $\Delta$ is closed.

( $\Longleftarrow$ ) Suppose $\Delta$ is closed in $X \times X$, so $(X \times X) \setminus \Delta$ is open. Let $x, y \in X$ with $x \neq y$. Then $(x, y) \in (X \times X) \setminus \Delta$. By definition of the product topology, there exists a basic open set $U \times V \subseteq X \times X$ such that $(x, y) \in U \times V \subseteq (X \times X) \setminus \Delta$. This implies $x \in U$ and $y \in V$. If there existed $z \in U \cap V$, then $(z, z) \in (U \times V) \cap \Delta$, contradicting $U \times V \subseteq (X \times X) \setminus \Delta$. Therefore $U \cap V = \varnothing$, so $X$ is Hausdorff. □

### Part (ii).

*Proof.* We check the three properties of a homeomorphism for $d \colon X \to \Delta$:

1. **Bijectivity:**
   - *Injective*: $d(x_1) = d(x_2) \implies (x_1, x_1) = (x_2, x_2) \implies x_1 = x_2$.
   - *Surjective*: Any element in $\Delta$ is of the form $(x, x) = d(x)$ for some $x \in X$.
2. **Continuity of $d$:** The component functions of $d$ are $d_1 = \mathrm{id}_X$ and $d_2 = \mathrm{id}_X$. Since the identity map $\mathrm{id}_X \colon X \to X$ is continuous, $d \colon X \to X \times X$ is continuous by the characteristic property of product spaces. Restricting the codomain to the subspace $\Delta$ preserves continuity.
3. **Continuity of $d^{-1}$:** The inverse map $d^{-1} \colon \Delta \to X$ is given by $d^{-1}(x, x) = x$. This is the restriction of the first coordinate projection $p_1 \colon X \times X \to X$ to $\Delta$. Since projection maps are continuous, its restriction $d^{-1} = p_1|_\Delta$ is continuous.

Thus, $d \colon X \to \Delta$ is a homeomorphism. □

## Problem 3.9

Let $X$ and $Y$ be topological spaces, and let $f, g \colon X \to Y$ be continuous maps. Assume that $Y$ is a Hausdorff space. Prove the following statements:

(i) The coincidence set $E = \{x \in X \mid f(x) = g(x)\}$ is closed in $X$.

(ii) If $D$ is a dense subset of $X$ and $f|_D = g|_D$, then $f = g$ on all of $X$.

(iii) The graph of $f$, defined as $\Gamma_f = \{(x, f(x)) \in X \times Y \mid x \in X\}$, is closed in the product space $X \times Y$.

(iv) Assume that $f \colon X \to Y$ is a continuous, open, and surjective map. Then the fiber relation set $R = \{(x_1, x_2) \in X \times X \mid f(x_1) = f(x_2)\}$ is closed in $X \times X$ if and only if $Y$ is Hausdorff.

### Solution

*submitted by* **Pula Karan**

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### Part (i).

*Proof.* Define $h \colon X \to Y \times Y$ by $h(x) = (f(x), g(x))$. Since $f$ and $g$ are continuous, $h$ is continuous. Because $Y$ is Hausdorff, the diagonal $\Delta_Y = \{(y, y) \in Y \times Y\}$ is closed in $Y \times Y$ by Problem 3.8(i). Notice that:

$$x \in E \iff f(x) = g(x) \iff (f(x), g(x)) \in \Delta_Y \iff x \in h^{-1}(\Delta_Y)$$

Thus $E = h^{-1}(\Delta_Y)$. Since $\Delta_Y$ is closed and $h$ is continuous, $E$ is closed in $X$. □

### Part (ii).

*Proof.* Consider the coincidence set $E = \{x \in X \mid f(x) = g(x)\}$. By part (i), $E$ is closed in $X$. Since $f|_D = g|_D$, we have $D \subseteq E$. Taking closures gives $\overline{D} \subseteq \overline{E} = E$. Since $D$ is dense in $X$, $\overline{D} = X$. Therefore $X \subseteq E$, which means $E = X$, so $f(x) = g(x)$ for all $x \in X$. □

**Part (iii).**

*Proof.* Define $F\colon X \times Y \to Y \times Y$ by $F(x,y) = (f(x), y)$. $F$ is continuous because its component maps $f \circ p_1$ and $p_2$ are continuous. Since $Y$ is Hausdorff, $\Delta_Y$ is closed in $Y \times Y$. Notice that:

$$(x,y) \in \Gamma_f \iff y = f(x) \iff (f(x), y) \in \Delta_Y \iff F(x,y) \in \Delta_Y$$

Thus $\Gamma_f = F^{-1}(\Delta_Y)$. Since $F$ is continuous and $\Delta_Y$ is closed, $\Gamma_f$ is closed in $X \times Y$. □

**Part (iv).**

*Proof.* Consider the product map $f \times f\colon X \times X \to Y \times Y$ defined by $(f \times f)(x_1, x_2) = (f(x_1), f(x_2))$. Since $f$ is continuous, open, and surjective, $f \times f$ is also continuous, open, and surjective. Notice that $R = (f \times f)^{-1}(\Delta_Y)$.

( $\Longleftarrow$ ) If $Y$ is Hausdorff, $\Delta_Y$ is closed in $Y \times Y$. By continuity of $f \times f$, $R = (f \times f)^{-1}(\Delta_Y)$ is closed in $X \times X$.

( $\Longrightarrow$ ) Suppose $R$ is closed in $X \times X$. Since $f \times f$ is surjective, $(f \times f)(R) = \Delta_Y$. Taking complements yields:

$$(Y \times Y) \setminus \Delta_Y = (f \times f)\big((X \times X) \setminus R\big)$$

Since $R$ is closed, $(X \times X) \setminus R$ is open. Because $f \times f$ is an open map, $(Y \times Y) \setminus \Delta_Y$ is open in $Y \times Y$, which means $\Delta_Y$ is closed in $Y \times Y$. By Problem 3.8(i), $Y$ is Hausdorff. □

## Problem 3.10

Let $X$ be a topological space and $A \subset X$ be dense in $X$. Let $Y$ be a Hausdorff space and $f\colon A \to Y$ be a continuous map. Show that if $f$ may be extended to a continuous map $g\colon X \to Y$, then $g$ is uniquely determined by $f$.

**Solution** *submitted by* **Pula Karan**

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*Proof.* Suppose $g_1, g_2\colon X \to Y$ are two continuous extensions of $f\colon A \to Y$, so $g_1|_A = f$ and $g_2|_A = f$.

1. Since $g_1|_A = g_2|_A = f$, the maps $g_1$ and $g_2$ agree on the subset $A \subseteq X$.
2. $A$ is dense in $X$, and $Y$ is a Hausdorff space.
3. By **Problem 3.9(ii)**, if two continuous functions from $X$ into a Hausdorff space $Y$ agree on a dense subset $A \subseteq X$, they must agree on all of $X$.

Therefore, $g_1(x) = g_2(x)$ for all $x \in X$, proving that the continuous extension $g$ is uniquely determined by $f$. □

# Problem Set 4

## Problem 4.1

Let $X$ and $Y$ be two topological spaces; and $a \in X$, $b \in Y$ be two given points. Prove that the maps $f\colon X \to X \times Y$ and $g\colon Y \to X \times Y$ given by

$$f(x) = (x, b) \quad \text{and} \quad g(y) = (a, y)$$

are embeddings.

**Solution pending**

## Problem 4.2

Suppose that $f, g\colon X \to \mathbb{R}$ are two continuous maps.

(i). Prove that the set $\{x \in X \mid f(x) \leq g(x)\}$ is closed in $X$.

(ii). Prove that the map $h\colon X \to \mathbb{R}$ given by $h(x) = \min\{f(x), g(x)\}$ is continuous.

**Solution pending**

## Problem 4.3

Let $\{A_\alpha\}$ be a collection of subsets of $X$ such that $X = \bigcup_\alpha A_\alpha$. Suppose that $f\colon X \to Y$ is a map such that the restriction map $f|_{A_\alpha}\colon A_\alpha \to Y$ is continuous for each $\alpha$.

(i). Prove that if the collection $\{A_\alpha\}$ is finite and each set $A_\alpha$ is closed, then $f$ is continuous.

(ii). Find an example where the collection $\{A_\alpha\}$ is countable and each set $A_\alpha$ is closed, but $f$ is not continuous.

(iii). An indexed family of sets $\{A_\alpha\}$ is said to be *locally finite* if each point $x$ has a neighborhood that intersects $A_\alpha$ for only finitely many values of $\alpha$. Prove that if the family $\{A_\alpha\}$ is locally finite and each set $A_\alpha$ is closed, then $f$ is continuous.

**Solution pending**

## Problem 4.4

Let $f\colon X \to A$ and $g\colon Y \to B$ be two continuous maps. Define $f \times g\colon X \times Y \to A \times B$ by $(f \times g)(x, y) = (f(x), g(y))$. Prove that $f \times g$ is a continuous map.

**Solution pending**

## Problem 4.5

Let $(X, d)$ be a metric space and let $S \subseteq X$ be a non-empty subset. For any $x \in X$, define the distance from $x$ to $S$ by $d(x, S) = \inf_{s \in S} d(x, s)$.

(1) Show that for any $x, y \in X$, $|d(x, S) - d(y, S)| \leq d(x, y)$.

(2) Conclude that $f_S \colon X \to \mathbb{R}$ defined by $f_S(x) = d(x, S)$ is continuous on $X$.

### Solution

*submitted by* **Aaditya**

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**Proof.**

(1) By the triangle inequality, for every $s \in S$,

$$d(x, s) \leq d(x, y) + d(y, s).$$

Taking the infimum over $s \in S$, we get

$$d(x, S) \leq d(x, y) + d(y, S).$$

Thus,

$$d(x, S) - d(y, S) \leq d(x, y).$$

Interchanging $x$ and $y$ gives

$$d(y, S) - d(x, S) \leq d(x, y).$$

Therefore,

$$\boxed{|d(x, S) - d(y, S)| \leq d(x, y).}$$

(2) Let $x_0 \in X$ and $\varepsilon > 0$. Choose $\delta = \varepsilon$.

If $d(x, x_0) < \delta$, then by (1),

$$\begin{aligned} |f_S(x) - f_S(x_0)| &= |d(x, S) - d(x_0, S)| \\ &\leq d(x, x_0) \\ &< \varepsilon. \end{aligned}$$

Hence, $f_S$ is continuous on $X$. □

## Problem 4.6

Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence of connected subspaces of $X$ such that $A_n \cap A_{n+1} \neq \emptyset$ for all $n \in \mathbb{N}$. Prove that $\bigcup_{n=1}^{\infty} A_n$ is connected.

### Solution

*submitted by* **Aaditya**

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**Proof.**

Let

$$A = \bigcup_{n=1}^{\infty} A_n.$$

Suppose, for contradiction, that $A$ is disconnected. Then

$$A = U \cup V,$$

where $U, V$ are non-empty, disjoint and open in $A$.

Since each $A_n$ is connected and

$$A_n = (A_n \cap U) \cup (A_n \cap V),$$

we must have

$$A_n \subseteq U \quad \text{or} \quad A_n \subseteq V.$$

Without loss of generality, let $A_1 \subseteq U$.

Since $A_1 \cap A_2 \neq \emptyset$, there exists $p \in A_1 \cap A_2$. Thus $p \in U$, so $A_2 \not\subseteq V$. Consequently, $A_2 \subseteq U$.

Repeating this argument inductively, we obtain

$$A_n \subseteq U \quad \text{for every } n \in \mathbb{N}.$$

Therefore,

$$A = \bigcup_{n=1}^{\infty} A_n \subseteq U,$$

contradicting $V \neq \emptyset$.

Hence,

$$\boxed{\bigcup_{n=1}^{\infty} A_n \text{ is connected.}}$$

□

## Problem 4.7

Determine whether or not $\mathbb{R}^\omega$ is connected under the uniform topology.

### Solution

*submitted by* **Aaditya**

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**Proof.**

The uniform metric is

$$\bar{\rho}(x, y) = \sup_n \min\{|x_n - y_n|, 1\}.$$

Define

$$\mathcal{B} = \left\{ x \in \mathbb{R}^\omega : \sup_n |x_n| < \infty \right\}$$

and

$$\mathcal{U} = \mathbb{R}^\omega \setminus \mathcal{B}.$$

We show that both sets are open.

First, observe that

$$\bar{\rho}(x, y) < 1 \implies |x_n - y_n| < 1 \quad \forall n.$$

**Claim 1: $\mathcal{B}$ is open.**

Let $x \in \mathcal{B}$. Then $|x_n| \leq M$ for some $M > 0$ and all $n$.

If $\bar{\rho}(x, y) < 1$, then

$$|y_n| \leq |x_n| + |y_n - x_n| < M + 1.$$

Hence $y \in \mathcal{B}$, proving that $\mathcal{B}$ is open.

**Claim 2: $\mathcal{U}$ is open.**

Let $x \in \mathcal{U}$ and suppose $\bar{\rho}(x, y) < 1$.

If $y$ were bounded, say $|y_n| \leq M$, then

$$|x_n| \leq |y_n| + |x_n - y_n| < M + 1.$$

This contradicts the fact that $x$ is unbounded.

Thus $y \in \mathcal{U}$, proving that $\mathcal{U}$ is open.

Finally, $\mathcal{B}$ and $\mathcal{U}$ are disjoint, non-empty, and

$$\mathcal{B} \cup \mathcal{U} = \mathbb{R}^\omega.$$

Therefore, they form a separation of $\mathbb{R}^\omega$. Hence,

$$\boxed{\mathbb{R}^\omega \text{ is not connected.}}$$

□

## Problem 4.8

Determine whether or not the lower limit topology $\mathbb{R}_\ell$ on $\mathbb{R}$ is connected.

### Solution

*submitted by* **Aaditya**

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**Proof.**

The lower limit topology has the basis

$$\mathcal{B} = \{[a, b) : a < b\}.$$

Consider

$$U = (-\infty, 0), \qquad V = [0, \infty).$$

We show that both sets are open.

If $x \in U$, then

$$x \in [x, 0) \subseteq U.$$

Since $[x, 0)$ is a basis element, $U$ is open.

Similarly, if $x \in V$, then

$$x \in [x, x + 1) \subseteq V.$$

Thus, $V$ is also open.

Moreover,

$$U \cap V = \emptyset, \qquad U \cup V = \mathbb{R}.$$

Both sets are non-empty. Therefore, $(U, V)$ forms a separation of $\mathbb{R}_\ell$.

Hence,

$$\boxed{\mathbb{R}_\ell \text{ is not connected.}}$$

□

### Problem 4.9

Let $C \subset X$ be a connected subspace of $X$ and $A \subset X$. Prove that if $C$ intersects both $A$ and $X \setminus A$, then $C$ intersects $\operatorname{Bd} A$.

**Solution pending**

### Problem 4.10

Let $X$ and $Y$ be two connected subspaces and $A \subsetneq X$ and $B \subsetneq Y$. Prove that $(X \times Y) \setminus (A \times B)$ is connected.

**Solution pending**

### Problem 4.11

Let $Y \subset X$; and let $X$ and $Y$ be connected. Prove that if $A$ and $B$ form a separation of $X \setminus Y$, then both $Y \cup A$ and $Y \cup B$ are connected.

**Solution pending**

# Problem Set 5

## Problem 5.1

Prove that the punctured Euclidean space $X = \mathbb{R}^n \setminus \{0\}$ is path connected if and only if $n > 1$.

**Solution pending**

## Problem 5.2

Prove that $\mathbb{R}^2 \setminus \mathbb{Q}^2$ with the subspace topology of $\mathbb{R}^2$ is connected, but $\mathbb{R} \setminus \mathbb{Q}$ with the subspace topology of $\mathbb{R}$ is not connected.

**Solution pending**

## Problem 5.3

Let $X$ and $Y$ be topological spaces, and suppose there exist embeddings $f\colon X \to Y$ and $g\colon Y \to X$. Is it necessary that $X$ and $Y$ are homeomorphic? Prove or provide a counterexample.

**Solution pending**

## Problem 5.4

Let $X$ be a non-empty topological space and let $\mathscr{B}$ be a base for the topology on $X$. Prove that $X$ is compact if and only if every open cover of $X$ consisting of elements of $\mathscr{B}$ has a finite subcover.

**Solution pending**

## Problem 5.5

Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on $X$ such that both $(X, \mathcal{T}_1)$ and $(X, \mathcal{T}_2)$ are compact Hausdorff spaces. Prove that either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable.

### Solution

*submitted by* **Prabhu**

*Proof.* Suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are comparable. Without loss of generality, let

$$\mathcal{T}_1 \subseteq \mathcal{T}_2.$$

Consider the identity map

$$I\colon (X, \mathcal{T}_2) \to (X, \mathcal{T}_1), \qquad I(x) := x.$$

Clearly, $I$ is *bijective*.

For all $U \in \mathcal{T}_1$, we have

$$I^{-1}(U) = U \in \mathcal{T}_2.$$

Hence, $I$ is *continuous*.

Since $I$ is a bijective continuous map from a compact space to a Hausdorff space, $I$ is a *homeomorphism*. So, $I^{-1}$ is also continuous.

Thus, for all $U \in \mathcal{T}_2$, we have

$$(I^{-1})^{-1}(U) = I(U) = U \in \mathcal{T}_1.$$

Hence,

$$\mathcal{T}_2 \subseteq \mathcal{T}_1.$$

Therefore,

$$\boxed{\mathcal{T}_1 = \mathcal{T}_2.}$$

Hence, either $\mathcal{T}_1 = \mathcal{T}_2$ or they are not comparable. □

## Problem 5.6

Prove that every compact subspace of a metric space is closed and bounded in that metric topology. Find an example of a metric space in which a closed and bounded subspace is not compact.

### Solution

*submitted by* **Prabhu**

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*Proof.* Let $(X, d)$ be a metric space and let $K \subseteq X$ be compact.

We know that every metric space is Hausdorff and a compact subspace of a Hausdorff space is closed. Since $X$ is a metric space it is Hausdorff. And $K$ is a compact subspace of $X$, thus it is *closed* in $X$.

Let $a \in X$. Consider the collection

$$\mathcal{U} := \{B(a, n) : n \in \mathbb{N}\},$$

where

$$B(a, n) := \{x \in X : d(x, a) < n\}.$$

For all $x \in K$, $d(x, a)$ is finite; so, $\exists\, n \in \mathbb{N}$ such that $d(x, a) < n$; i.e., $x \in B(a, n)$. Thus, $\mathcal{U}$ is an open cover of $A$.

Since $K$ is compact, there exists a finite subcover $\{B(a, n_1), \ldots, B(a, n_m)\}$. Therefore,

$$K \subseteq \bigcup_{i=1}^{m} B(a, n_i).$$

Let $N := \max\{n_1, n_2, \ldots, n_m\}$. Since $B(a, n_i) \subseteq B(a, N)$ for every $i$, we get

$$K \subseteq \bigcup_{i=1}^{m} B(a, n_i) \subseteq B(a, N).$$

Therefore, $K$ is *bounded*. Hence,

$$\boxed{K \text{ is closed and bounded.}}$$

□

**Example:** Consider the metric space $X := (0, 1)$ with the usual metric ($\because d(x, y) := |x - y|$), and let $A := X$. Clearly, $A$ is closed ($\because$ it is the entire space) and bounded ($\because \forall x \in A,\ 0 < x < 1$) in $X$.

However, consider the collection

$$\mathcal{U} = \left\{ \left( \frac{1}{n}, 1 \right) : n \in \mathbb{N},\ n \geq 2 \right\}.$$

Each member of $\mathcal{U}$ is open in the subspace $(0, 1)$. And,

$$(0, 1) = \bigcup_{n=2}^{\infty} \left( \frac{1}{n}, 1 \right).$$

Thus, $\mathcal{U}$ is an open cover of $A$.

For every finite subcollection $\left\{ \left( \frac{1}{n_1}, 1 \right), \left( \frac{1}{n_2}, 1 \right), \ldots, \left( \frac{1}{n_k}, 1 \right) \right\}$, let $N = \max\{n_1, n_2, \ldots, n_k\}$. Since $n_i \leq N$ for every $i = 1, \ldots, k$, we have

$$\left( \frac{1}{n_i}, 1 \right) \subseteq \left( \frac{1}{N}, 1 \right).$$

Therefore,

$$\bigcup_{i=1}^{k} \left( \frac{1}{n_i}, 1 \right) \subseteq \left( \frac{1}{N}, 1 \right) \subsetneq (0, 1).$$

It cannot cover $(0, 1)$, since points in $(0, 1/N]$ remain uncovered. Hence, $\mathcal{U}$ has no finite subcover. Therefore,

$$\boxed{(0, 1) \text{ is closed and bounded but not compact in } (0, 1).}$$

## Problem 5.7

Let $X$ be a compact metric space. Prove that if $f \colon X \to X$ is an isometry (distance preserving map) of $X$, then $f$ is a homeomorphism.

**Solution pending**

## Problem 5.8

Let $\mathbb{R}$ be given the cocountable topology. Is the closed interval $[0, 1]$ a compact subspace of $\mathbb{R}$ under this topology?

**Solution pending**

### Problem 5.9

Let $f : X \to Y$ be a continuous map between topological spaces. Show that if $X$ is compact and $Y$ is Hausdorff, then $f$ is a closed map.

**Solution pending**

### Problem 5.10

Prove that if $Y$ is compact, then the projection $\pi_X : X \times Y \to X$ is a closed map.

**Solution pending**

# Problem Set 6

## Problem 6.1

Let $p\colon X \twoheadrightarrow Y$ be a closed continuous surjective map such that $p^{-1}(y)$ is compact for each $y \in Y$. Prove that if $Y$ is compact, then so is $X$.

**Solution pending**

## Problem 6.2

Let $X$ be a compact metric space. Prove that if $f\colon X \to X$ is an isometry (distance preserving map) of $X$, then $f$ is a homeomorphism.

**Solution pending**

## Problem 6.3

Prove that the composition of two quotient maps is a quotient map.

**Solution pending**

## Problem 6.4

Prove that a surjective continuous map $f\colon X \twoheadrightarrow Y$ that is either an open map or a closed map is a quotient map.

**Solution pending**

## Problem 6.5

Prove that a bijective quotient map is a homeomorphism.

**Solution pending**

### Problem 6.6

Let $f_1 \colon X_1 \to Y_1$ and $f_2 \colon X_2 \to Y_2$ be maps between topological spaces, and let $f_1 \times f_2 \colon X_1 \times X_2 \to Y_1 \times Y_2$ denote the product map $(x_1, x_2) \mapsto (f_1(x_1), f_2(x_2))$.

(a) Prove that if $f_1$ and $f_2$ are continuous, then $f_1 \times f_2$ is continuous with respect to the product topologies.

(b) Give a counterexample to show that if $p_1 \colon X_1 \to Y_1$ and $p_2 \colon X_2 \to Y_2$ are quotient maps, $p_1 \times p_2$ need not be a quotient map.

**Solution pending**

### Problem 6.7

Let $X$ and $Y$ be topological spaces, and $f \colon X \to Y$ be a bijective continuous map. Show that if $Y$ is Hausdorff, then $X$ is also Hausdorff.

**Solution pending**

### Problem 6.8

Let $X$ be a compact topological space and $p \colon X \twoheadrightarrow Y$ be a quotient map.

(a) Prove that $Y$ is compact.

(b) Show that if $Y$ is Hausdorff, then $p$ is a closed map.

**Solution pending**

### Problem 6.9

Let $p \colon X \to Y$ be a continuous surjection. Prove that $p$ is a closed map if and only if for every subset $S \subseteq Y$ and every open set $U \subseteq X$ containing $p^{-1}(S)$, there exists an open set $V \subseteq Y$ containing $S$ such that $p^{-1}(V) \subseteq U$.

**Solution pending**

# Problem Set 7 (Munkres §22)

## Problem 7.1

Check the details of Example 3.[a]

[a] Example 3 of §22 (The Quotient Topology) in J. R. Munkres, *Topology*, 2nd ed.

**Solution pending**

## Problem 7.2

(a) Let $p\colon X \to Y$ be a continuous map. Show that if there is a continuous map $f\colon Y \to X$ such that $p \circ f$ equals the identity map of $Y$, then $p$ is a quotient map.

(b) If $A \subset X$, a *retraction* of $X$ onto $A$ is a continuous map $r\colon X \to A$ such that $r(a) = a$ for each $a \in A$. Show that a retraction is a quotient map.

**Solution pending**

## Problem 7.3

Let $\pi_1\colon \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be projection on the first coordinate. Let $A$ be the subspace of $\mathbb{R} \times \mathbb{R}$ consisting of all points $x \times y$ for which either $x \geq 0$ or $y = 0$ (or both); let $q\colon A \to \mathbb{R}$ be obtained by restricting $\pi_1$. Show that $q$ is a quotient map that is neither open nor closed.

**Solution pending**

## Problem 7.4

(a) Define an equivalence relation on the plane $X = \mathbb{R}^2$ as follows:

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0 + y_0^2 = x_1 + y_1^2.$$

Let $X^*$ be the corresponding quotient space. It is homeomorphic to a familiar space; what is it? [*Hint:* Set $g(x \times y) = x + y^2$.]

(b) Repeat (a) for the equivalence relation

$$x_0 \times y_0 \sim x_1 \times y_1 \quad \text{if} \quad x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

**Solution pending**

## Problem 7.5

Let $p\colon X \to Y$ be an open map. Show that if $A$ is open in $X$, then the map $q\colon A \to p(A)$ obtained by restricting $p$ is an open map.

**Solution pending**

## Problem 7.6

Recall that $\mathbb{R}_K$ denotes the real line in the $K$-topology (see §13). Let $Y$ be the quotient space obtained from $\mathbb{R}_K$ by collapsing the set $K$ to a point; let $p\colon \mathbb{R}_K \to Y$ be the quotient map.

(a) Show that $Y$ satisfies the $T_1$ axiom, but is not Hausdorff.

(b) Show that $p \times p\colon \mathbb{R}_K \times \mathbb{R}_K \to Y \times Y$ is not a quotient map. [*Hint:* The diagonal is not closed in $Y \times Y$, but its inverse image is closed in $\mathbb{R}_K \times \mathbb{R}_K$.]

**Solution pending**

# Problem Set 8 (Munkres §51, §52)

## Problem 8.1

*(§51, Ex. 1)* Show that if $h, h' \colon X \to Y$ are homotopic and $k, k' \colon Y \to Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.

**Solution pending**

## Problem 8.2

*(§51, Ex. 2)* Given spaces $X$ and $Y$, let $[X, Y]$ denote the set of homotopy classes of maps of $X$ into $Y$.

(a) Let $I = [0, 1]$. Show that for any $X$, the set $[X, I]$ has a single element.

(b) Show that if $Y$ is path connected, the set $[I, Y]$ has a single element.

**Solution pending**

## Problem 8.3

*(§51, Ex. 3)* A space $X$ is said to be *contractible* if the identity map $i_X \colon X \to X$ is nulhomotopic.

(a) Show that $I$ and $\mathbb{R}$ are contractible.

(b) Show that a contractible space is path connected.

(c) Show that if $Y$ is contractible, then for any $X$, the set $[X, Y]$ has a single element.

(d) Show that if $X$ is contractible and $Y$ is path connected, then $[X, Y]$ has a single element.

**Solution pending**

## Problem 8.4

*(§52, Ex. 1)* A subset $A$ of $\mathbb{R}^n$ is said to be *star convex* if for some point $a_0$ of $A$, all the line segments joining $a_0$ to other points of $A$ lie in $A$.

(a) Find a star convex set that is not convex.

(b) Show that if $A$ is star convex, $A$ is simply connected.

**Solution pending**

## Problem 8.5

*(§52, Ex. 2)* Let $\alpha$ be a path in $X$ from $x_0$ to $x_1$; let $\beta$ be a path in $X$ from $x_1$ to $x_2$. Show that if $\gamma = \alpha * \beta$, then $\hat{\gamma} = \hat{\beta} \circ \hat{\alpha}$.

**Solution pending**

## Problem 8.6

*(§52, Ex. 3)* Let $x_0$ and $x_1$ be points of the path-connected space $X$. Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair $\alpha$ and $\beta$ of paths from $x_0$ to $x_1$, we have $\hat{\alpha} = \hat{\beta}$.

**Solution pending**

## Problem 8.7

*(§52, Ex. 4)* Let $A \subset X$; suppose $r\colon X \to A$ is a continuous map such that $r(a) = a$ for each $a \in A$. (The map $r$ is called a *retraction* of $X$ onto $A$.) If $a_0 \in A$, show that

$$r_*\colon \pi_1(X, a_0) \longrightarrow \pi_1(A, a_0)$$

is surjective.

**Solution pending**

## Problem 8.8

*(§52, Ex. 5)* Let $A$ be a subspace of $\mathbb{R}^n$; let $h\colon (A, a_0) \to (Y, y_0)$. Show that if $h$ is extendable to a continuous map of $\mathbb{R}^n$ into $Y$, then $h_*$ is the trivial homomorphism (the homomorphism that maps everything to the identity element).

**Solution pending**

## Problem 8.9

*(§52, Ex. 6)* Show that if $X$ is path connected, the homomorphism induced by a continuous map is independent of base point, up to isomorphisms of the groups involved. More precisely, let $h\colon X \to Y$ be continuous, with $h(x_0) = y_0$ and $h(x_1) = y_1$. Let $\alpha$ be a path in $X$ from $x_0$ to $x_1$, and let $\beta = h \circ \alpha$. Show that

$$\hat{\beta} \circ (h_{x_0})_* = (h_{x_1})_* \circ \hat{\alpha}.$$

This equation expresses the fact that the following diagram of maps "commutes."

$$\begin{array}{ccc} \pi_1(X, x_0) & \xrightarrow{(h_{x_0})_*} & \pi_1(Y, y_0) \\ \hat{\alpha}\downarrow & & \downarrow\hat{\beta} \\ \pi_1(X, x_1) & \xrightarrow[(h_{x_1})_*]{} & \pi_1(Y, y_1) \end{array}$$

**Solution pending**

## Problem 8.10

*(§52, Ex. 7)* Let $G$ be a topological group with operation $\cdot$ and identity element $x_0$. Let $\Omega(G, x_0)$ denote the set of all loops in $G$ based at $x_0$. If $f, g \in \Omega(G, x_0)$, let us define a loop $f \otimes g$ by the rule

$$(f \otimes g)(s) = f(s) \cdot g(s).$$

(a) Show that this operation makes the set $\Omega(G, x_0)$ into a group.

(b) Show that this operation induces a group operation $\otimes$ on $\pi_1(G, x_0)$.

(c) Show that the two group operations $*$ and $\otimes$ on $\pi_1(G, x_0)$ are the same. [*Hint:* Compute $(f * e_{x_0}) \otimes (e_{x_0} * g)$.]

(d) Show that $\pi_1(G, x_0)$ is abelian.

**Solution pending**